



SCOPE OF ACCREDITATION TO ISO/IEC 17025:2005
& ANSI/NCSL Z540-1-1994

APPLIED CALIBRATION SERVICES, LLC
 2662 Forest Hills Loop Road
 Suite D
 Wilson, NC 27893
 Brandon Owen Phone: 252-237-CAL1 (2251)

CALIBRATION

Valid To: February 28, 2017

Certificate Number: 3374.01

In recognition of the successful completion of the A2LA evaluation process, accreditation is granted to this laboratory to perform the following calibrations¹:

I. Electrical – DC/Low Frequency

Parameter/Equipment	Range	CMC ² (±)	Comments
Resistance – Generate	(0 to 10.9999) Ω (11 to 32.9999) Ω (33 to 109.9999) Ω (110 to 329.9999) Ω 330 Ω to 1.099999 kΩ (1.1 to 3.299999) kΩ (3.3 to 10.99999) kΩ (11 to 32.99999) kΩ (33 to 109.9999) kΩ (110 to 329.99999) kΩ 330 kΩ to 1.099999 MΩ (1.1 to 3.299999) MΩ (3.3 to 10.99999) MΩ (11 to 32.99999) MΩ (33 to 109.9999) MΩ (110 to 329.9999) MΩ (320 to 1100) MΩ	40 μΩ/Ω + 0.001 Ω 30 μΩ/Ω + 0.0015 Ω 28 μΩ/Ω + 0.0014 Ω 28 μΩ/Ω + 0.002 Ω 28 μΩ/Ω + 0.002 Ω 28 μΩ/Ω + 0.02 Ω 28 μΩ/Ω + 0.02 Ω 28 μΩ/Ω + 0.2 Ω 28 μΩ/Ω + 0.2 Ω 32 μΩ/Ω + 2 Ω 32 μΩ/Ω + 7 Ω 60 μΩ/Ω + 50 Ω 130 μΩ/Ω + 70 Ω 250 μΩ/Ω + 2.7 kΩ 500 μΩ/Ω + 3.2 kΩ 3 mΩ/Ω + 100 kΩ 15 mΩ/Ω + 510 kΩ	Multifunction calibrator
DC Voltage – Generate	(0 to 329.9999) mV 330 mV to 3.299999 V (3.3 to 32.99999) V (33 to 329.9999) V (330 to 1020) V	20 μV/V + 1 μV 11 μV/V + 2 μV 12 μV/V + 20 μV 18 μV/V + 150 μV 18 μV/V + 1.5 mV	Multifunction calibrator

Parameter/Equipment	Range	CMC ² (±)	Comments
DC Current – Generate	(0 to 329.999) μ A 330 μ A to 3.29999 mA (3.3 to 32.9999) mA (33 to 329.9999) mA 330 mA to 1.09999 A (1.1 to 2.99999) A (3 to 10.9999) A (11 to 20.5) A	150 μ A/A + 0.02 μ A 100 μ A/A + 0.05 μ A 100 μ A/A + 0.25 μ A 100 μ A/A + 2.5 μ A 200 μ A/A + 40 μ A 380 μ A/A + 40 μ A 500 μ A/A + 500 μ A 1 mA/A + 750 μ A	Multifunction calibrator

Parameter/Range	Frequency	CMC ² (±)	Comments
AC Voltage – Generate			
(1 to 32.999) mV	(10 to 45) Hz 45 Hz to 10 kHz (10 to 20) kHz (20 to 50) kHz (50 to 100) kHz (100 to 500) kHz	800 μ V/V + 6 μ V 150 μ V/V + 6 μ V 200 μ V/V + 6 μ V 1 mV/V + 6 μ V 3.5 mV/V + 12 μ V 8 mV/V + 50 μ V	Multifunction calibrator
(33 to 329.999) mV	(10 to 45) Hz 45 Hz to 10 kHz (10 to 20) kHz (20 to 50) kHz (50 to 100) kHz (100 to 500) kHz	300 μ V/V + 8 μ V 145 μ V/V + 8 μ V 160 μ V/V + 8 μ V 350 μ V/V + 8 μ V 800 μ V/V + 32 μ V 2 mV/V + 70 μ V	
(0.33 to 3.29999) V	(10 to 45) Hz 45 Hz to 10 kHz (10 to 20) kHz (20 to 50) kHz (50 to 100) kHz (100 to 500) kHz	300 μ V/V + 50 μ V 150 μ V/V + 60 μ V 190 μ V/V + 60 μ V 300 μ V/V + 50 μ V 700 μ V/V + 130 μ V 2.4 mV/V + 600 μ V	
(3.3 to 32.9999) V	(10 to 45) Hz 45 Hz to 10 kHz (10 to 20) kHz (20 to 50) kHz (50 to 100) kHz	300 μ V/V + 650 μ V 150 μ V/V + 600 μ V 240 μ V/V + 600 μ V 350 μ V/V + 600 μ V 900 μ V/V + 1.6 mV	
(33 to 329.999) V	45 Hz to 1 kHz (1 to 10) kHz (10 to 20) kHz (20 to 50) kHz (50 to 100) kHz	190 μ V/V + 2 mV 200 μ V/V + 6 mV 250 μ V/V + 6 mV 300 μ V/V + 6 mV 2 mV/V + 50 mV	
(330 to 1020) V	45 Hz to 1 kHz (1 to 5) kHz (5 to 10) kHz	300 μ V/V + 10 mV 250 μ V/V + 10 mV 300 μ V/V + 10 mV	

Parameter/Range	Frequency	CMC ^{2, 5} (±)	Comments
AC Current – Generate			
(29 to 329.99) uA	(10 to 20) Hz (20 to 45) Hz 45 Hz to 1 kHz (1 to 5) kHz (5 to 10) kHz (10 to 30) kHz	0.2 % + 0.1 µA 0.15 % + 0.1 µA 0.13 % + 0.1 µA 0.3 % + 0.15 µA 0.8 % + 0.2 µA 1.6 % + 0.4 µA	Multifunction calibrator
(0.33 to 3.29999) mA	(10 to 20) Hz (20 to 45) Hz 45 Hz to 1 kHz (1 to 5) kHz (5 to 10) kHz (10 to 30) kHz	0.2 % + 0.15 µA 0.15 % + 0.15 µA 0.1 % + 0.15 µA 0.2 % + 0.2 µA 0.5 % + 0.3 µA 1 % + 0.6 µA	
(3.3 to 32.9999) mA	(10 to 20) Hz (20 to 45) Hz 45 Hz to 1 kHz (1 to 5) kHz (5 to 10) kHz (10 to 30) kHz	0.18 % + 2 µA 0.09 % + 2 µA 0.04 % + 2 µA 0.08 % + 2 µA 0.2 % + 3 µA 0.4 % + 4 µA	
(33 to 329.999) mA	(10 to 20) Hz (20 to 45) Hz 45 Hz to 1 kHz (1 to 5) kHz (5 to 10) kHz (10 to 30) kHz	0.18 % + 20 µA 0.09 % + 20 µA 0.04 % + 20 µA 0.10 % + 50 µA 0.2 % + 100 µA 0.4 % + 200 µA	
(0.33 to 1.09999) A	(10 to 45) Hz 45 Hz to 1 kHz (1 to 5) kHz (5 to 10) kHz	0.18 % + 100 µA 0.05 % + 100 µA 0.6 % + 1 mA 2.5 % + 5 mA	
(1.1 to 2.999999) A	(10 to 45) Hz 45 Hz to 1 kHz (1 to 5) kHz (5 to 10) kHz	0.18 % + 100 µA 0.06 % + 100 µA 0.6 % + 1 mA 2.5 % + 5 mA	
(3 to 10.9999) A	(45 to 100) Hz (0.1 to 1) kHz (1 to 5) kHz	0.06 % + 2 mA 0.1 % + 2 mA 3 % + 2 mA	
(11 to 20.5) A	(45 to 100) Hz (0.1 to 1) kHz (1 to 5) kHz	0.12 % + 5 mA 0.15 % + 5 mA 3 % + 5 mA	

Parameter/Equipment	Range	CMC ^{2, 5} (±)	Comments
Capacitance	(220 to 399.9) pF (0.4 to 1.09999) nF (1.1 to 3.29999) nF (3.3 to 10.9999) nF (11 to 32.999) nF (0.033 to 109.999) nF (110 to 329.999) nF (0.33 to 1.0999) µF (1.1 to 3.29999) µF (3.3 to 10.9999) µF (11 to 32.9999) µF (33 to 109.9999) µF (110 to 329.999) µF (0.33 to 1.09999) mF (1.1 to 3.29999) mF (3.3 to 10.9999) mF (11 to 32.9999) mF (33 to 110) mF	0.5 % + 10 pF 0.5 % + 0.1 nF 0.5 % + 0.01 nF 0.25 % + 0.01 nF 0.25 % + 0.01 nF 0.25 % + 0.01 nF 0.25 % + 0.03 nF 0.25 % + 1 nF 0.25 % + 3 nF 0.25 % + 10 nF 0.40 % + 30 nF 0.45 % + 0.1 µF 0.45 % + 0.3 µF 0.45 % + 1 µF 0.45 % + 3 µF 0.45 % + 10 µF 0.75 % + 30 µF 1.1 % + 0.1 mF	Multifunction calibrator
Thermocouple – Generate / Measure			
Type J	(-210 to -100) °C (-100 to -30) °C (-30 to 150) °C (150 to 760) °C (760 to 1200) °C	0.27 °C 0.16 °C 0.14 °C 0.17 °C 0.23 °C	Multifunction calibrator
Type K	(-200 to -100) °C (-100 to -25) °C (-25 to 120) °C (120 to 1000) °C (1000 to 1372) °C	0.33 °C 0.18 °C 0.16 °C 0.26 °C 0.40 °C	
Type S	(0 to 250) °C (250 to 1000) °C (1000 to 1400) °C (1400 to 1767) °C	0.47 °C 0.36 °C 0.37 °C 0.46 °C	
Type T	(-250 to -150) °C (-150 to 0) °C (0 to 120) °C (120 to 400) °C	0.63 °C 0.24 °C 0.16 °C 0.14 °C	

Parameter/Equipment	Range	CMC ² (±)	Comments
Resistance – Measure	(0 to 12) Ω (12 to 120) Ω (120 to 1200) Ω (1.2 to 12) kΩ (12 to 120) kΩ 120 kΩ to 1.2 MΩ (1.2 to 12) MΩ (12 to 120) MΩ	20 μΩ/Ω + 59 μΩ 16 μΩ/Ω + 590 μΩ 14 μΩ/Ω + 590 μΩ 14 μΩ/Ω + 6 mΩ 14 μΩ/Ω + 60 mΩ 19 μΩ/Ω + 2.7 Ω 58 μΩ/Ω + 120 Ω 580 μΩ/Ω + 1.2 kΩ	8 ½ digit DMM
DC Voltage – Measure	(0 to 120) mV 120 mV to 1.2V (1.2 to 12) V (12 to 120) V (120 to 1050) V	13 μV/V + 0.38 μV 10 μV/V + 0.38 μV 10 μV/V + 0.59 μV 13 μV/V + 37 μV 20 μV/V + 120 μV	8 ½ digit DMM
DC Current – Measure	(0 to 120) nA (0.12 to 1.2) uA (1.2 to 12) uA (12 to 120) uA (0.12 to 1.2) mA (1.2 to 12) mA (12 to 120) mA (0.12 to 1.2) A	35 μA/A + 47 pA 23 μA/A + 47 pA 23 μA/A + 120 pA 23 μA/A + 0.94 nA 23 μA/A + 6 nA 23 μA/A + 60 nA 41 μA/A + 0.6 μA 130 μA/A + 12 μA	8 ½ digit DMM

Parameter/Range	Frequency	CMC ^{2, 5} (±)	Comments
AC Voltage – Measure (0 to 12) mV	(1 to 40) Hz 40 Hz to 1 kHz (1 to 20) kHz (20 to 50) kHz (50 to 100) kHz (100 to 300) kHz	0.035 % + 5.3 μV 0.023 % + 4.2 μV 0.035 % + 4.3 μV 0.12 % + 9.5 μV 0.58 % + 9.5 μV 4.6 % + 66 μV	8 ½ digit DMM
(12 to 120) mV	(1 to 40) Hz 40 Hz to 1 kHz (1 to 20) kHz (20 to 50) kHz (50 to 100) kHz (100 to 300) kHz 300 kHz to 1 MHz (1 to 2) MHz	0.009 % + 7.3 μV 0.009 % + 8.1 μV 0.017 % + 8.1 μV 0.035 % + 37 μV 0.092 % + 37 μV 0.35 % + 70 μV 1.2 % + 70 μV 1.7 % + 70 μV	

Parameter/Range	Frequency	CMC ^{2,5} (±)	Comments
AC Voltage – Measure			
(0.12 to 1.2) V	(1 to 40) Hz 40 Hz to 1 kHz (1 to 20) kHz (20 to 50) kHz (50 to 100) kHz (100 to 300) kHz 300 kHz to 1 MHz (1 to 2) MHz	0.009 % + 68 µV 0.009 % + 55 µV 0.017 % + 73 µV 0.035 % + 130 µV 0.092 % + 210 µV 0.35 % + 610 µV 1.2 % + 1.7 mV 1.7 % + 1.7 mV	8 ½ digit DMM
(1.2 to 12) V	(1 to 40) Hz 40 Hz to 1 kHz (1 to 20) kHz (20 to 50) kHz (50 to 100) kHz (100 to 300) kHz 300 kHz to 1 MHz (1 to 2) MHz	0.009 % + 700 µV 0.009 % + 540 µV 0.017 % + 720 µV 0.035 % + 1.3 mV 0.092 % + 1.6 mV 0.35 % + 5.2 mV 1.2 % + 19 mV 1.7 % + 19 mV	
(12 to 120) V	(1 to 40) Hz 40 Hz to 1 kHz (1 to 20) kHz (20 to 50) kHz (50 to 100) kHz (100 to 300) kHz 300 kHz to 1 MHz	0.023 % + 8.4 mV 0.023 % + 8.4 mV 0.023 % + 12 mV 0.041 % + 14 mV 0.14 % + 34 mV 0.46 % + 34 mV 1.7 % + 34 mV	
(120 to 1000) V	(1 to 40) Hz 40 Hz to 1 kHz (1 to 20) kHz (20 to 50) kHz (50 to 100) kHz	0.05 % + 77 mV 0.05 % + 77 mV 0.07 % + 77 mV 0.14 % + 77 mV 0.35 % + 77 mV	
AC Current – Measure			
(0 to 120) µA	(10 to 20) Hz (20 to 45) Hz (45 to 100) Hz (0.1 to 1) kHz	0.46 % + 47 nA 0.17 % + 47 nA 0.07 % + 47 nA 0.07 % + 47 nA	8 ½ digit DMM
(0.12 to 1.2) mA	(10 to 20) Hz (20 to 45) Hz (45 to 100) Hz (0.1 to 5) kHz (5 to 20) kHz (20 to 50) kHz (50 to 100) kHz	0.46 % + 260 nA 0.17 % + 260 nA 0.07 % + 260 nA 0.04 % + 260 nA 0.07 % + 260 nA 0.46 % + 480 nA 0.64 % + 1.7 µA	

Parameter/Range	Frequency	CMC ^{2, 5} ($\pm$)	Comments
AC Current – Measure			
(1.2 to 12) mA	(10 to 20) Hz (20 to 45) Hz (45 to 100) Hz (0.1 to 5) kHz (5 to 20) kHz (20 to 50) kHz (50 to 100) kHz	0.46 % + 3 μ A 0.17 % + 3 μ A 0.07 % + 3 μ A 0.04 % + 3 μ A 0.07 % + 3 μ A 0.46 % + 5 μ A 0.64 % + 17 μ A	8 ½ digit DMM
(12 to 120) mA	(10 to 20) Hz (20 to 45) Hz (45 to 100) Hz (0.1 to 5) kHz (5 to 20) kHz (20 to 50) kHz (50 to 100) kHz	0.46 % + 30 μ A 0.17 % + 30 μ A 0.07 % + 30 μ A 0.04 % + 30 μ A 0.07 % + 30 μ A 0.46 % + 50 μ A 0.64 % + 170 μ A	
(0.12 to 1) A	(10 to 20) Hz (20 to 45) Hz (45 to 100) Hz (0.1 to 5) kHz (5 to 20) kHz (20 to 50) kHz	0.46 % + 2.1 mA 0.19 % + 2.1 mA 0.1 % + 2.1 mA 0.12 % + 2.1 mA 0.35 % + 2.1 mA 1.2 % + 2.2 mA	

II. Mechanical

Parameter/Equipment	Range	CMC ^{2, 4, 5} ($\pm$)	Comments
Scales & Balances ³	(0 to 5) g	0.041 mg + 0.6R	ASTM Class 1 weights
	(0 to 200) g	0.29 mg + 0.6R	
Pipettes ³	(1 to 49) μ L (50 to 200) μ L 200 μ L to 5 mL	0.6 % + 0.00037 μ L/ μ L 0.28 μ L + 0.00037 μ L/ μ L 0.43 μ L + 0.00037 μ L/ μ L	ISO 8655

Parameter/Equipment	Range	CMC ^{2, 5} (±)	Comments
Pressure – Measure			
Absolute / Gauge	(0 to 30) psi (30 to 100) psi (0 to 300) psi (300 to 1000) psi	0.00072 psi 0.008 % 0.0072 psi 0.008 %	Pressure calibrator
Differential Pressure	(0 to 1) inch H ₂ O (1 to 10) inch H ₂ O (0 to 3) inch H ₂ O (3 to 30) inch H ₂ O	0.00084 inch H ₂ O 0.011 % + 0.00083 inch H ₂ O 0.0025 inch H ₂ O 0.011 % + 0.0025 inch H ₂ O	Low pressure calibrator
Torque Tools	(4 to 50) lbf·in (30 to 400) lbf·in (80 to 1000) lbf·ft (25 to 250) lbf·ft	0.32 % 0.29 % 0.27 % 0.29 %	Torque transducer calibration system

III. Thermodynamics

Parameter/Equipment	Range	CMC ^{2, 5} (±)	Comments
Temperature – Measure	(-80 to 0) °C 0.01 °C (TPW) (0 to 250) °C (250 to 550) °C (550 to 1000) °C (1000 to 1450) °C (-10 to 70) °C	0.012 °C 0.007 °C 0.014 °C 0.033 °C 1.2 % 2 % 0.06 °C	SPRT Type S thermocouple Environmental chamber
Humidity	(10 to 95) % RH	0.5 % RH	Environmental chamber

¹ This laboratory offers commercial calibration service and field calibration service.

² Calibration and Measurement Capability Uncertainty (CMC) is the smallest uncertainty of measurement that a laboratory can achieve within its scope of accreditation when performing more or less routine calibrations of nearly ideal measurement standards or nearly ideal measuring equipment. CMCs represent expanded uncertainties expressed at approximately the 95 % level of confidence, usually using a coverage factor of $k = 2$. The actual measurement uncertainty of a specific calibration performed by the laboratory may be greater than the CMC due to the behavior of the customer's device and to influences from the circumstances of the specific calibration.

³ Field calibration service is available for this calibration and this laboratory meets A2LA R104 – *General Requirements: Accreditation of Field Testing and Field Calibration Laboratories* for these calibrations. Please note the actual measurement uncertainties achievable on a customer's site can normally be expected to be larger than the CMC found on the A2LA Scope. Allowance must be made for aspects such as the environment at the place of calibration and for other possible adverse effects such as those caused by transportation of the calibration equipment. The usual allowance for the actual uncertainty introduced by the item being calibrated, (e.g. resolution) must also be considered and this, on its own, could result in the actual measurement uncertainty achievable on a customer's site being larger than the CMC.

⁴ R is the resolution.

⁵ In the statement of CMC, uncertainties expressed as a percent are to be read as percent of reading unless otherwise indicated.



Accredited Laboratory

A2LA has accredited

APPLIED CALIBRATION SERVICES, LLC

Wilson, NC

for technical competence in the field of

Calibration

This laboratory is accredited in accordance with the recognized International Standard ISO/IEC 17025:2005 *General requirements for the competence of testing and calibration laboratories*. This laboratory also meets the requirements of ANSI/NCSL Z540-1-1994 and R205 – Specific Requirements: Calibration Laboratory Accreditation Program. This accreditation demonstrates technical competence for a defined scope and the operation of a laboratory quality management system (*refer to joint ISO-ILAC-IAF Communiqué dated 8 January 2009*).



Presented this 6th day of July, 2015.

A blue ink signature of the President and CEO, written over a horizontal line.

President and CEO
For the Accreditation Council
Certificate Number 3374.01
Valid to February 28, 2017
Revised on November 25, 2016

For the calibrations to which this accreditation applies, please refer to the laboratory's Calibration Scope of Accreditation.